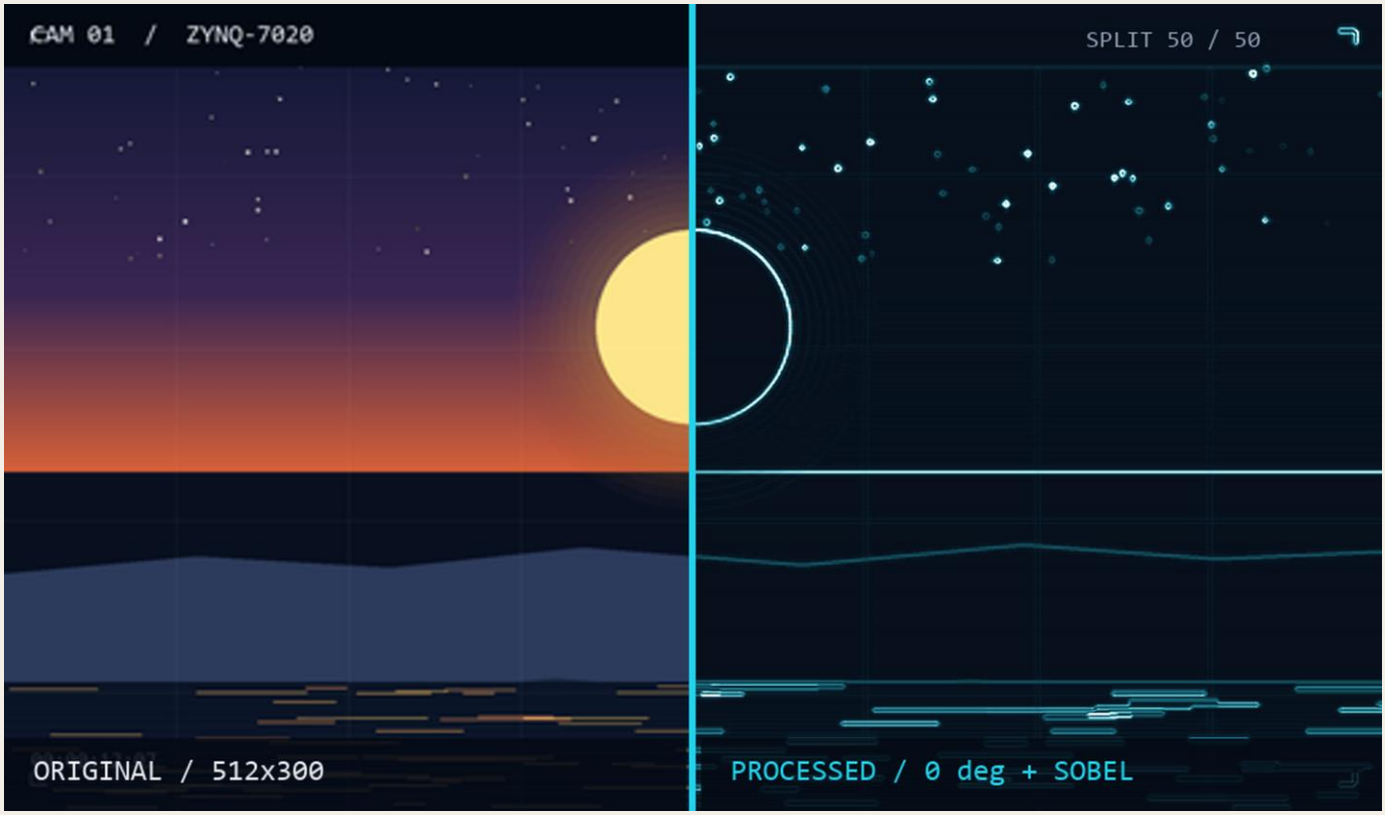
ZYNQ-7020 · A FULLY HARDWARE VIDEO NODE

One frame.
Two truths.

Original and processed pixels leave one frame buffer in the same clock, so the split screen compares content rather than screenshots. The node reports what it did on the wire at 1 Hz.



Rendered from the same 512×300 test frame through the implemented Sobel chain at 0°, no rotation.

01 SAME-FRAME A/B

A fixed 15-cycle effect chain sits behind both halves. The seam is a pixel decision, not a compositing trick.

02 CAUSE-BEARING TELEMETRY

Frames, aborts, missing rows, PL latency, pane integrity and frame gaps arrive as a 1 Hz UDP beacon, not just FPS.

03 CALIBRATED BY FAULT INJECTION

Inject 1-in-100 packet loss: aborts and missed rows must move, FPS must fall, recovery must return to baseline.

THE NUMBERS · ROUTED BUILD

1.341 ns	0.025 ns	70.00%	15 cycles
setup WNS · 0 failing endpoints	hold WHS · tightest path	BRAM · 98 of 140 tiles	fixed effect latency

RGMII → MAC → UDP → DDR → GEOMETRY → 6-STAGE EFFECTS → A/B SEAM → HDMI

512×300 RGB565 at 30 fps → 1024×600 at 59.5 Hz · xc7z020clg484-2 · 1 GbE UDP ingest · SD and test-pattern fallback

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